

Year	Return	Deviation from Mean	Deviation Squared
10	4.0%	0.0100	0.000100
Mean	3.0%		
Sum			0.005750

The standard deviation is the square root of the sum of the squared deviations, divided by $n - 1$:

$$s = \sqrt{\frac{0.005750}{(10 - 1)}} = 2.5276\%.$$

8. B is correct. The target semideviation of the returns over the 10 years with a target of 2 percent is calculated as follows:

Year	Return	Deviation Squared below Target of 2%
1	4.5%	
2	6.0%	
3	1.5%	0.000025
4	-2.0%	0.001600
5	0.0%	0.000400
6	4.5%	
7	3.5%	
8	2.5%	
9	5.5%	
10	4.0%	
Sum		0.002025

The target semideviation is the square root of the sum of the squared deviations from the target, divided by $n - 1$:

$$s_{\text{Target}} = \sqrt{\frac{0.002025}{(10 - 1)}} = 1.5\%.$$

9. B is correct. The CV is the ratio of the standard deviation to the mean, where a higher CV implies greater risk per unit of return.

$$CV_{\text{UTIL}} = \frac{s}{\bar{X}} = \frac{1.23\%}{2.10\%} = 0.59,$$

$$CV_{\text{MATR}} = \frac{s}{\bar{X}} = \frac{1.35\%}{1.25\%} = 1.08,$$

$$CV_{\text{INDU}} = \frac{s}{\bar{X}} = \frac{1.52\%}{3.01\%} = 0.51.$$